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Agentic AI in Cybersecurity Review of Autonomous Threat Detection and Adaptive Defense Mechanisms

[1] Dr. Sheily Panwar, [2] Habiba Abdelrahman

[1.] Assistant professor (Computer science/ Cyber Security), CUQ-Ulster University, Qatar

[2.] Student (B. Eng-AI), CUQ-Ulster University, Qatar

[1.] sheily.verma@cuq-ulster.edu.qa [2.] Abdelrahman-h@ulster.ac.uk

Abstract **–Regarding cybersecurity, this work explores Agentic AI's ability to independently identify risks, adapt to changing hazards, and enhance defensive systems. This work aims to investigate in the field of digital ecosystems security its ideas, practical applications, challenges, and possible future directions of research. Examining the most recent advancements in Agentic AI, their place fit into security measures systems, and how they differ from more traditional AI-driven security models by means of a comprehensive literature study revealed Part of the research are intelligent threat hunting, responsive risk reduction, self-learning security models, or autonomous threat detecting. Important concerns noted also are aggressive attacks, moral conundrums, artificial intelligence bias, and legal ramifications. The results lead agentic artificial intelligence to either give real-time anomaly detection, predictive threat assessment, or automated incident response, thereby enhancing cybersecurity resilience. It distinguishes itself from conventional artificial intelligence in its capacity to change in real-time to meet lately growing cyber dangers. This concurrently helps it to lower response times and stop APTs and zero-day vulnerabilities. Important impediments to its general implementation, meantime, are ethical concerns, explainability difficulties, and poor legislation. Since agentic artificial intelligence offers autonomous defense systems proactive against advanced threats, it changes the paradigm in cybersecurity. Even if it enhances security operations, responsible and efficient deployment calls for overcoming ethical, legal, and technical limitations. Future research should focus on developing more strong cybersecurity frameworks by combining Agentic AI with developing technologies as blockchain, federated learning, and quantum security**.

Keywords- Agentic AI, Cybersecurity, Autonomous Threat Detection, Self-Learning Models and Adaptive Risk Mitigation

I. INTRODUCTION

From others, agentic artificial intelligence systems specify autonomous decisions, adaptability, and proactive solution-finding free from constant human participation. Unlike classical AI, which often follows given rules or rely on supervised learning, agentic artificial intelligence is able to autonomously rate events, learn from changing risks, and intelligately reply to real-time security-related issues. Agentic artificial intelligence is crucial in the field of security, where the terrain is always shifting due to modern cyberthreats because enhancing electronic systems by means of recognition, reduction, and avoidance of attacks with lowest delay.[1]–[4]. Traditional cybersecurity solutions including rule-based platforms, detection utilising signatures, or human-driven reaction mechanisms struggle against APTs, zero-day holes, or adversarial AI attacks even if they perform adequately against known threats. Agentic artificial intelligence assists cybersecurity systems grow more resilient and

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responsive by automatically recognising anomalies, modifying security measures, projecting new assaults, and acting pro-actively to resist them. The major objective of this study is to investigate how agentic artificial intelligence's capabilities increase threat identification, incident response, and risk minimising while addressing related concerns including rivalry, ethical questions, and rules implication. It also tries to grasp the objectives and implications of agentic AI in security.[5]–[8]. This work aims to provide a comprehensive historical analysis of machine learning (AI) in cybersecurity with an eye on the change from more standard AI models to more brilliant or independent agentic AI-driven solutions.



Fig. 1 Agentic AI in cybersecurity[9]

The analysis will explore the key components of an agentic artificial intelligence in security: self-learning privacy models, governed response to events, bright threat hunting, or adaptive risk lowering techniques[10]–[15]. Furthermore stressed will be numerous practical uses like zero-day attack prediction, created intelligence-powered Security Operations Centres (SOCs), real-time virus and phishing surveillance, and artificial intelligence-driven systems for intrusion prevention and detection. Among the key issues with artificial intelligence biases, adversarial abuses, data privacy, and a shortage of explainability in machine learning techniques will be discussed in the review. Furthermore crucial is the need of laws to ensure responsible usage of artificial intelligence. With an eye towards accountability and decision openness, the conversation will also explore ethical issues about the proper blend of human oversight and total autonomy. The study will culminate with looking ahead to possible trends and directions of future investigation. To enhance decentralised cybersecurity, one such field is the mix of agentic artificial intelligence with modern technologies such blockchain, quantum security, and federated learning. The sophistication of cyberattacks is driving demand for clever, self-sufficient defensive systems; agentic artificial intelligence offers a fascinating new direction for cybersecurity evolution. This paper seeks to present a thorough knowledge of how agentic artificial intelligence is changing the field of cybersecurity and what actions are required to guarantee its moral and effective application in protection of digital environments. It tackles the opportunities as well as the difficulties related to the technology.

II. Related work

John 2025 et.al Large language models (LLMs) and generative artificial intelligence (GEA) help agentic artificial intelligence (AGR) to progressively enable autonomous systems. Although agentic artificial intelligence predates contemporary LLMs, their interaction with generative artificial intelligence has greatly enlarged their scope, capacity, and related hazards. The first of a series of guidelines from the OWASP Agentic Security Initiative (ASI), this paper offers a threat-model-based reference to newly arising agentic risks and addresses mitigating strategies[1].

Maleville 2024 et.al explores the potential of Key Enabling Technologies to change the way the EU deals with its expanding security threats. The report uses foresight analysis to evaluate AI, advanced sensing, blockchain, and drones, showcasing their potential to strengthen critical infrastructure resilience, combat crime and terrorism, and enhance law enforcement. It also exposes vulnerabilities, such as the possibility of criminal abuse or regulatory gaps, to these technologies. Strategic investments in safe creative thinking, legal harmonisation, and addressing social resilience—including legal harmonisation and legal harmony—are among the EU measures that should be prioritised in light of the results, since they will help with technology transformation while also reducing risks. This document provides practical insights on emerging security concerns, supporting a strong, safe, and resilient Europe. It complements the goals of the Commission for 2024-2020. Promising technological solutions that strike a balance between security needs and ethical and societal values, the recommendations aim to promote effective public-private partnerships, guarantee regulatory coherence across Member States, and reiterate the EU's leadership in sustainable and innovative policies in internal security.[16].

Tallam 2024 et.al in this age of unprecedented digital complexity, the cybersecurity landscape is evolving at a dizzying pace, challenging long-held beliefs about defence. The existence of Advanced Persistent Threats (APTs) emphasises the urgent need for adaptable strategies that seamlessly integrate human knowledge with cutting-edge AI tools in order to reveal the gaps in conventional security measures. By rethinking frameworks like the cyber death chain, improving threat intelligence processes, and integrating strong ethical

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governance into automated reaction systems, this paper investigates how the convergence of agentic AI and Frontier AI is changing cybersecurity. To construct a robust and adaptable defensive ecosystem, we examine the functions of real-time monitoring, autonomous incident response, and lifetime learning using prospective perspectives and real-world data. The study's overarching objective is to find a way to balance strict moral oversight with rapid technological advancement in order to build AI-driven security solutions that effectively combat emerging cyber threats without compromising the core human values of transparency, accountability, and fairness.[17].

Barua 2024 et.al Autonomous AI agents using huge language models may have undeniable societal benefits, but they suffer security threats from adversaries that necessitate swift defensive actions due to trust and safety concerns. Research on real-world robustness should prioritise many-shot jailbreaking and misleading alignment, two of the most advanced threats that cannot be mitigated by the static guardrails used in supervised training. In a dynamic multi-agent system, the dangers cannot be protected by static guardrails alone. Improving security for LLM-based agents is our goal. To do this, we have developed new evaluation frameworks that can identify potential threats and provide solutions to make operational deployments safe. Using GEMINI 1.5 pro and llama-3.3-70B as benchmarks, our work develops an anti-jailbreaking system. We employ tool-mediated adversarial scenarios with three examination techniques, including a Reverse Turing Test, to identify rogue agents. Finally, we use multi-agent simulations to analyse deceptive alignment. While GEMINI 1.5 pro's detection skills are good (94% accuracy), the system is vulnerable to protracted attacks due to the fact that diversity metrics are useless for prediction and prompt length raises attack success rates (ASR), exposing multiple complicated defects in the system. The findings highlight the necessity for adaptable security systems that combine agent-led active monitoring with system administrator-led flexible interventions, as existing models have the potential to introduce vulnerabilities that can result in an unreliable and susceptible system. Therefore, we want to address such situations and provide a comprehensive strategy to address the security issues in our work.[18].

Bengio 2024 et.al address how present AI training techniques lead to these hazards. Various situations and tests have indeed shown the possibility of artificial intelligence agents acting in dishonesty or following objectives not stated by human operators and that contradict human needs, including self-preservation. Applying the precautionary principle, we clearly see a great need for better, yet still practical substitutes for the present agency-driven course. Thus, we suggest as a fundamental building element for next developments of a trustworthy and secure by design non-agentic artificial intelligence system, which we name Scientist AI. Unlike acting in the world to either please or replicate people, this system is meant to explain it from observations. It consists of a question-answering inference machine and a world model producing theories to interpret facts. Both parts work with an explicit awareness of uncertainty to reduce the possibility of too optimistic forecasts. Given these factors, a Scientist AI could be employed to support human researchers in advancing scientific advancement especially in AI safety. Specifically, our method can be used as a guardrail against artificial intelligence agents developed despite the associated hazards. In the end, emphasising non-agentic artificial intelligence might help to avoid the hazards connected with the present direction while also allowing the advantages of AI progress. These reasons should inspire legislators, developers, and academics to support this safer road forward[3].

TABLE.1 LITERATURE SUMMARY

Study/Author	Objective	Methodology	Key Findings	Limitations & Future Scope
Wang 2025 [2]	CyberMentor improves cybersecurity education for non-traditional students by offering AI-driven, individualized guidance, which helps with information acquisition, improvement of abilities, and career preparation.	CyberMentor is an online tool that provides individualized cybersecurity training by combining RAG with agentic workflows and LangChain assessment.	CyberMentor is an AI-powered platform that helps students with their studying, career preparation, and skill development.	If CyberMentor wants to develop, it needs more testing, more domains, and more real-time adaptation.

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Humphreys 2025 [19]	Finding a middle ground between responsible business practices, innovation, and cybersecurity threats is essential for ethical AI implementation.	Ethical principles, cybersecurity threats, and procedures for responsible AI application were examined.	Ethical use of generative AI reduces cybersecurity concerns, disinformation, and over-reliance.	Adaptive ethical safeguards, legal frameworks, and the ever-changing dangers posed by AI should be the focus of future studies.
Malatji 2024 [20]	The AICD Framework is a set of guidelines for dealing with AI cybersecurity issues in an ethical, multidisciplinary, and adaptive manner.	Developed the AICD Foundation for AI cybersecurity and performed textual analyzes and literature assessments.	The AICD Framework emphasizes multidisciplinary approaches to AI cybersecurity, adaptive defenses, and ethical concerns.	The AICD Framework, new AI security risks, and interdisciplinary cooperation should all be the focus of future studies.
Access 2024 [21]	By automating routine tasks, improving threat detection, and providing more accurate predictions, AI and ML are enhancing cybersecurity.	Researched AI-driven threat detection using natural language processing, advanced learning, cases, and automation.	AI and ML improve cybersecurity by detecting threats, extracting intelligence, automating processes, and making predictions.	Integrating quantum cybersecurity, data privacy, AI breakthroughs, and ethical considerations should all be the focus of future research.
Adenekan 2023 [22]	AI and GRNN improve autonomous network management, threat detection, IoT security, and AIOps.	Protected the Internet of Things using artificial intelligence, GRNN, ML, anomaly detection, or predictive analytics.	Together, GRNN and AI make networks more secure, optimize IoT devices, and pave the way for self-healing systems.	Scalability, ethics, and ever-changing dangers are some of the obstacles; adaptability is an area that needs further investigation in the future.

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III. Background and Evolution of AI in Cybersecurity

Firewalls, antivirus programs, rule-based security mechanisms, and intrusion detection systems (IDS) have long been the backbone of cybersecurity. These conventional security solutions locate and mitigate threats by using preexisting signatures and heuristics. Advanced persistent threats (APTs), polymorphic malware, zero-day vulnerabilities, and other new and complex attacks are difficult to defend against using these methods, even while they work well against known attack patterns. Rather than being proactive, rule-based security solutions rely on threat databases that are constantly being updated. Additionally, human security responses are becoming insufficient in real-time threat identification and mitigation due to hackers' increasingly complex and automated attack methods. Traditional security measures are unable to deal with emerging cyber risks, and the rapid expansion of digital infrastructure, cloud computing, and the IoT has only made matters worse. Cybersecurity has increasingly applied artificial intelligence and machine learning approaches to go above these restrictions and improve detection, analysis, and response systems. Using machine learning (ML) algorithms to sort through vast volumes of data, identify anomalies, and discover dangerous trends considerably more rapidly and precisely than human analysts, artificial intelligence (AI) driven cybersecurity solutions By spotting events of non-standard behaviour, machine learning models—which combine supervised and unsupervised learning approaches—help cybersecurity systems to find hitherto undetectable hazards. These developments have made cybersecurity defences significantly more efficient, which opens the path for predictive threat detection, automated threat intelligence, and real-time incident response. Early artificial intelligence cybersecurity solutions, however, largely served as decision-support systems, depending on pre-defined training data and human supervision to tune responses. These AI-driven models needed continuous human supervision to retrain the models and update the security policies even if they raised detection rates and response times. Cybersecurity suffers much-reaching effects from the change from conventional rule-based artificial intelligence to agentic AI. Agentic artificial intelligence that can learn and adapt on its own, therefore enabling autonomous security judgements, marks a new era in cybersecurity defence tactics.[23]–[28]. Unlike conventional artificial intelligence, which depends on pre-defined models, agentic artificial intelligence develops independently in response to changing threat scenarios and real-time security measure adjustments help to reduce new threats. Without human involvement, these AI-powered entities use neural networks, deep learning, and reinforcement learning to continuously improve their threat intelligence capabilities. Organisations can use autonomous security agents—that is, agentic artificial intelligence—that can instantly identify, evaluate, and remove cyberthreats in cybersecurity systems. Artificial intelligence, intelligent threat-hunting technologies, and self-healing networks managed by security operations centres (SOCs) have all helped them to more actively defend against cyberattacks with less human control. Thanks in part to predictive analytics—which help agentic artificial intelligence—cysecurity systems are now able to foresee attack tendencies and act preventatively. Although agentic artificial intelligence has the ability to transform cybersecurity, it also brings fresh issues including the requirement of legislation guaranteeing responsible AI deployment, ethical questions about autonomous decision-making, and adversarial assaults aiming at AI models. Agentic artificial intelligence is becoming ever more important in contemporary cybersecurity systems since it provides a more strong and flexible protection against developing cyberthreats.

IV. Core Components of Agentic AI in Cybersecurity

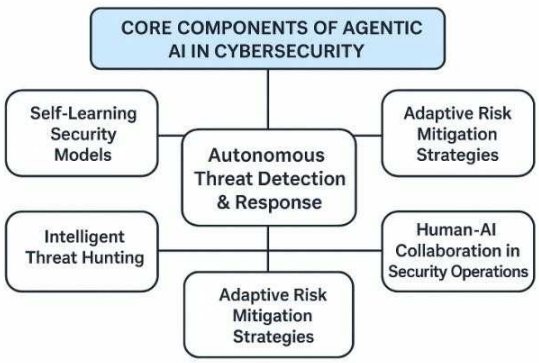


Fig. 2 Core component of Agentic AI in cybersecurity (compiled by Researcher)

A. Autonomous Threat Detection & Response

Cybersecurity systems can now detect and respond to cyber threats automatically, without human interaction, in real time. Conventional systems are vulnerable to new types of attacks because they use detection based on signatures and established criteria. Anomaly detection, behavioural analysis, and machine learning are the tools that agentic AI uses to dynamically identify questionable actions. Isolating infected systems, blocking harmful traffic, or deploying countermeasures are all examples of automatic responses that AI-driven security mechanisms might launch upon threat detection. By taking preventative measures, reaction times are decreased, damage is minimised, and cybersecurity infrastructures are strengthened to withstand advanced threats, such as

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zero-day exploits and advanced persistent threats (APTs).[29].

B. Self-Learning Security Models

Self-learning security models empower AI-driven cybersecurity systems to evolve and adapt continuously without requiring constant human oversight. Unlike traditional machine learning models that depend on periodic retraining, agentic AI employs reinforcement learning and neural networks to refine its understanding of emerging threats. These models analyze patterns from vast datasets, identify deviations from normal behavior, and autonomously update their threat intelligence. This capacity helps cybersecurity systems to keep ahead of cybercrime schemes, therefore ensuring that security protocols remain useful even when attackers create new approaches. By enabling self-improvement, self-learning security models dramatically raise threat detection or incident response capabilities.

C. Adaptive Risk Mitigation Strategies

One of the adaptive risk decreasing solutions is real-time risk assessment based dynamic adjustment of security systems. To always evaluate risk, agentic artificial intelligence investigates many threat indicators like network behaviour, user activity, as well as outside threat intelligence. Unlike static security systems, which may grow obsolete as cyber threats change, adaptive technology powered by AI automatically change security policies, distribute resources, and use countermeasures in reaction to new risks. This guarantees best protection and helps to maintain system performance. Adaptive risk reducing solutions increase general security resilience and reduce likely attack surfaces by giving high-risk risks top attention and correcting holes before they are employed.

D. Intelligent Threat Hunting

Using intelligent threat hunting—AI-driven automation—a company's network can be actively searched for hidden risks. Unlike conventional threat detection, agentic artificial intelligence-driven threat hunting uses predictive analytics, anomaly detection, and natural language processing (NLP) to identify alarming trends before they become full-scale attacks. These technologies could give insightful analysis when security teams review massive volumes of security data and link indications of compromise (IoCs). Intelligent threat hunting regularly searches for new and growing dangers, therefore strengthening a company's defences against insider threats, sophisticated malware campaigns, and covert assaults.

E. Human-AI Collaboration in Security Operations

By combining human wisdom and judgement with artificial intelligence's speed and efficiency, people and machines can improve security operations.[30]. Agentic artificial intelligence can handle basic security chores, but human analysts offer strategic decision-making, ethical control, and vital contextual knowledge. By filtering noise, prioritising warnings, and recommending reaction actions, AI-driven solutions help security teams to free analysts to concentrate on difficult risks needing human judgement. AI and human knowledge working together improves incident response effectiveness, lowers false positives, and strengthens general cybersecurity strategy. Combining human intelligence with automation will help companies to have a more balanced and successful cyber defence.

V. Applications of Agentic AI in Cybersecurity

- **Intrusion Detection & Prevention Systems (IDPS):** Agentic artificial intelligence enhances IDPS in several aspects including automated detection of anomalies, network traffic analysis for patterns, and real-time threat mitigating. This is accomplished without reference to set signatures or guidelines.
- **Real-time Malware & Phishing Detection:** By analysing enormous databases, seeing harmful trends, and instantaneously blocking malware and phishing attempts, AI-driven models instantly lower human error and strengthen cybersecurity defences.
- **Zero-Day Attack Prediction & Mitigation:** Agentic artificial intelligence uses behavioural anomalies, proactive patch deployment, and countermeasures before attackers take use of undisclosed system weaknesses to forecast zero-day vulnerabilities.
- **Automated Incident Response and Recovery:** AI-driven security solutions save downtime and improve resilience against cyberattacks by independently spotting breaches, including threats, and running recovery procedures.
- **AI-driven SOC (Security Operations Centers):** Through alert filtering, threat prioritising, and real-time threat intelligence assistance to analysts, artificial intelligence maximises SOCs for faster, more efficient cybersecurity management.
- **Deception Technology (AI-created honeypots, traps for cybercriminals):** Artificial intelligence-generated honey pots and trickery entice attackers into under control situations, gathering data and reducing risk before actual systems are attacked [31]–[35].

VI. Challenges & Future Trends

Challenges & Ethical Considerations

- **Bias in AI-driven cybersecurity solutions:** AI algorithms' natural biases in the training data could lead to erroneous threat assessments or

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uneven security responses. Reaching equity calls for diverse training providers and unbiased datasets.

- **Adversarial Attacks Against AI Models:** Strong combative defence mechanisms are required to stop security failures driven by artificial intelligence since cybercriminals exploit AI vulnerabilities by modifying inputs to mislead detection systems.
- **Explainability & Transparency in Automated Decision-Making:** Cybersecurity decisions motivated by artificial intelligence should be rational to guarantee responsibility. Black-box models bring hazards, thus explainable AI is rather necessary for regulatory compliance and confidence development.
- **Data Privacy & Compliance Issues:** Among the problems that surface when intelligence based safety systems search massive amounts of sensitive data in line with global standards including GDPR are data privacy, law regulation, and ethical data management.
- **Human Oversight vs. Full Autonomy Risks:** Although automation improves cybersecurity efficacy, too depending too much on computations without human input could have negative effects and calls for a compromise between autonomy and human judgement.

Future Trends & Research Directions

Technology for Security Agentic artificial intelligence moving forward will be determined by AI-augmented cyberdefense strategies—where flexible AI models proactive observe or neutralise threats before they escalate. More complicated attack and defence systems resulting from the rise of AI-powered cyberwarfare will depend on continuous antagonistic AI developments. Integration with particle security or blockchain will enhance encryption and data honesty and therefore ensuring strong protection against emerging risks.[36]–[41]. Federated learning will encourage distributed cybersecurity, thereby allowing discreetly threatening intelligence exchange across businesses. Moreover mostly dependent on the evolution of governance and regulatory systems is the creation of ethical, open, responsible security protocols powered by artificial intelligence.

VII. Conclusion

In the process conducting an analysis of its evolution, fundamental components, real-world applications, and related obstacles, this review has thoroughly investigated the developing field of Agentic AI in cybersecurity, thereby attaining its aims. Agentic artificial intelligence offers a paradigm change in how companies defend against always changing cyber threats by substituting self-adaptive, intelligent frameworks for conventional rule-based systems. Five key components—autonomous threat detection and response, self-learning models, adaptive risk mitigating, intelligent threat hunting, and human-AI cooperation—which taken together enable cybersecurity systems to run actively and autonomously—were emphasised in the review. Agentic artificial intelligence's pragmatic worth in reducing challenging risks such zero-day vulnerabilities and sophisticated persistent attacks was shown by real-world use cases. Simultaneously, the study recognised the ethical, technical, and operational issues that had to be resolved including data privacy issues, lack of explainability, and artificial intelligence bias. The results highlight how Agentic artificial intelligence greatly improves cybersecurity resilience and reactivity, but its efficacy depends on careful application and ongoing monitoring. Finally, this review provides a basic knowledge of how Agentic AI can turn cybersecurity into a dynamic, intelligent, and ethical defence ecosystem and prepares the ground for next developments combining blockchain, federated learning, and quantum computing to create a strong and safe digital future.

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